WORKED SOLUTION

STEP I 2017, Question 2

Pure • Inequalities

STEP I 2017, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q2

The key idea

Integration preserves inequalities: if $f(t) \leq g(t)$ on an interval $[a, b]$ with $a < b$, then $\int_a^b f \leq \int_a^b g$. The only care needed is with the direction of integration, because when $x < 1$ the interval $[1, x]$ runs backwards and the inequality flips. In the final part the same idea is used again: integrating a non-negative function from 1 to y gives something with the same sign as $y - 1$, and that sign is exactly what we need to divide through safely.

Solution

Part (i)

Case $x \geq 1$. For $1 \leq t \leq x$ we have $\frac{1}{t} \leq 1$, and the interval runs forwards, so

$$\int_1^x \frac{1}{t}\,\mathrm{d}t \leq \int_1^x 1\,\mathrm{d}t, \qquad \text{that is} \qquad \ln x \leq x - 1.$$

Case $0 < x \leq 1$. Now we integrate over $x \leq t \leq 1$, where $t \leq 1$ and so $\frac{1}{t} \geq 1$. Then

$$\int_x^1 \frac{1}{t}\,\mathrm{d}t \geq \int_x^1 1\,\mathrm{d}t, \qquad \text{that is} \qquad -\ln x \geq 1 - x.$$

Multiplying by -1 reverses the inequality and gives $\ln x \leq x - 1$ again. So $(*)$ holds for all $x > 0$.

Part (ii)

Case $x \geq 1$. For $t \geq 1$ we have $\frac{1}{t^2} \leq \frac{1}{t}$, so

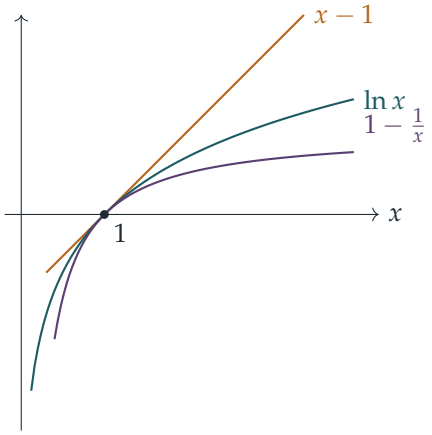
$$\int_1^x \frac{1}{t^2}\,\mathrm{d}t \leq \int_1^x \frac{1}{t}\,\mathrm{d}t, \qquad \text{that is} \qquad 1 - \frac{1}{x} \leq \ln x.$$

Case $0 < x \leq 1$. For $x \leq t \leq 1$ the inequality reverses, $\frac{1}{t^2} \geq \frac{1}{t}$ (multiply $t \leq 1$ by the positive number $\frac{1}{t^2}$). Integrating over $[x, 1]$,

$$\frac{1}{x} - 1 \geq -\ln x,$$

and multiplying by -1 gives $\ln x \geq 1 - \frac{1}{x}$. So $(**)$ holds for all $x > 0$.

(Alternatively, $(**)$ follows from $(*)$ at once: put $1/x$ in place of x in $(*)$ to get $-\ln x \leq \frac{1}{x} - 1$.)



All three curves touch at $x = 1$, where each has gradient 1; $\ln x$ is squeezed between the other two.

Part (iii)

Both $(*)$ and $(**)$ say that a certain function is non-negative on $x > 0$:

$$f(x) = x - 1 - \ln x \geq 0, \qquad g(x) = \ln x - 1 + \frac{1}{x} \geq 0.$$

We need a simple fact about integrating such a function.

Ask yourself If $h(x) \geq 0$ for all $x > 0$, what is the sign of $\int_1^y h(x)\,\mathrm{d}x$ when $y < 1$?

If $y > 1$ the integral is over a forward interval, so it is ≥ 0. If $0 < y < 1$, then $\int_1^y h = -\int_y^1 h \leq 0$. In either case

$$(y - 1)\int_1^y h(x)\,\mathrm{d}x \;\geq\; 0.$$

This lets us handle both $y > 1$ and $0 < y < 1$ together.

We need $\int \ln x \,\mathrm{d}x = x\ln x - x$ (integrating by parts with $u = \ln x$), so

$$\int_1^y \ln x \,\mathrm{d}x = y\ln y - y + 1.$$

The right-hand inequality. Integrating f,

$$\begin{aligned}\int_1^y f(x)\,\mathrm{d}x &= \frac{(y-1)^2}{2} - (y\ln y - y + 1) = \frac{(y-1)^2}{2} + (y-1) - y\ln y \\ &= \frac{(y-1)(y+1)}{2} - y\ln y.\end{aligned}$$

By the fact above,

$$(y-1)\left[\frac{(y-1)(y+1)}{2} - y\ln y\right] \geq 0 \quad\Longrightarrow\quad \frac{(y-1)^2(y+1)}{2} \geq y(y-1)\ln y.$$

Since $y > 0$ and $y \neq 1$, the number $y(y-1)^2$ is positive, and dividing by it keeps the direction:

$$\frac{y+1}{2y} \geq \frac{\ln y}{y-1}.$$

The left-hand inequality. Integrating g,

$$\int_1^y g(x)\,\mathrm{d}x = (y\ln y - y + 1) - (y-1) + \ln y = (y+1)\ln y - 2(y-1).$$

By the fact above,

$$(y-1)[(y+1)\ln y - 2(y-1)] \geq 0 \quad\Longrightarrow\quad (y+1)(y-1)\ln y \geq 2(y-1)^2.$$

Now divide by $(y+1)(y-1)^2 > 0$:

$$\frac{\ln y}{y-1} \geq \frac{2}{y+1}.$$

Putting the two together,

$$\frac{2}{y+1} \leq \frac{\ln y}{y-1} \leq \frac{y+1}{2y} \qquad \text{for } y > 0,\ y \neq 1,$$

as required.

Ask yourself Why multiply by $y-1$ before dividing, rather than dividing straight away?

Dividing an inequality by $y-1$ would reverse it when $y < 1$, forcing two separate cases. Multiplying by $y-1$ first produces an expression containing $(y-1)^2$, which is positive whichever side of 1 we are on, so one argument covers both cases.

Checks

- At $y = 2$: $\frac{2}{3} \approx 0.667 \leq \ln 2 \approx 0.693 \leq \frac{3}{4} = 0.75$.
- At $y = \frac{1}{2}$: $\frac{4}{3} \approx 1.333 \leq 2\ln 2 \approx 1.386 \leq \frac{3}{2}$.
- As $y \to 1$ all three expressions tend to 1 (the middle one is the gradient of ln at 1), consistent with the curves touching in the diagram.
- Replacing y by $1/y$ and then dividing through by y gives back exactly the same pair of inequalities, so the result for $0 < y < 1$ is equivalent to the result for $y > 1$.

Common slips

- Integrating over $[1, x]$ when $x < 1$ without noticing that the inequality reverses.
- Dividing by $y - 1$ without considering its sign, and so proving part (iii) only for $y > 1$.

Beyond STEP

The middle quantity $\dfrac{\ln y}{y-1}$ is the reciprocal of the *logarithmic mean* of 1 and y, so the left-hand inequality says that the logarithmic mean is at most the arithmetic mean $\frac{1+y}{2}$. In fact the logarithmic mean is also at least the geometric mean $\sqrt{y}$, which gives $\frac{\ln y}{y-1} \leq \frac{1}{\sqrt{y}}$, a sharper upper bound than $\frac{y+1}{2y}$. The technique of integrating an inequality, and the monotonicity of the integral that justifies it, are basic tools in a first analysis course. Abbott's *Understanding Analysis* treats the integral carefully.

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